INCH-POUND

MIL-P-48698 (AR)
AMENDMENT 6
01 September 1998
SUPERSEDING
AMENDMENT 5
13 September 1996

MILITARY SPECIFICATION

PROBE AND S&A ASSEMBLY, TOW-2/E6

This Amendment forms a part of Military Specification MIL-P-48698 (AR), dated 27 April 1987, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.11 Add the following:

"MIL-STD-1169 - Packaging, Packing and Marking for shipment of inert Ammunition components

Delete:

"DOD-STD-2000 - Soldering Technology - High Quality/High Reliability"

Substitute:

"DOD-STD-2000-4 - General purpose soldering requirements for electrical and electronic equipment."

AMSC N/A FSC 1336

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PAGE 3

3.4, delete paragraph in its entirety and substitute the following paragraph:

"3.4 Safety and arming device. The Safety and Arming device (S&A shall conform to all requirements of MIL-S-14799. The S&A shall be in the safe condition prior to the subsequent assembly of the probe. The red plug shall be evident inside the base terminal board to indicate the piston actuator has not been fired. The rotor shaft shall be in the proper orientation (See Figure 9). The gap between the spring detent and the bottom of the G sensing leaf slot shall not exceed one (1) times the wire diameter of the spring detent (See Figure 9). A gap that is equal to or exceeds two (2) times the wire diameter of the detent spring shall be classified as a "Critical" defect. A gap greater than one(1) times the wire diameter but less than two (2)times The wire diameter of the detent spring shall be classified as a "Major" defect. If any Safety and Arming Device fails the critical inspection or the major inspection, notify the Government technical agency within two (2) working days to determine the deposition of the device."

3.6: Line 3: Delete "is".

PAGE 4

3.17: Delete paragraph in its entirety and substitute with:

"3.17 Open circuit test. The probe shall exhibit an open circuit resistance greater than or equal to 1.0 megohms across the tip switch and the ogive switch, and a circuit resistance of greater than or equal to 900k ohms but less than or equal to 30 megohms across the detonator switch in either the stowed or deployed position. Detonator resistance between 100 and 100k ohms shall be classified as "critical" defect. Detonator resistance less than 100 ohms or greater than 100k ohms but less than 900k ohms shall be classified as "Major" defect. Detonator resistance greater than 30 megohms shall be considered a "no test"."

3.18: Delete paragraph in its entirety and substitute with:

"3.18 Continuity circuit test. The probe shall exhibit a resistance less than or equal to 1.0 ohms for the tip ground circuit, base S&A actuator circuit, ogive switch ground circuit and ogive switch live circuit in either the deployed or stowed position.

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The stowed probe shall exhibit a resistance of less than or equal to 1.0 ohms for the tip actuator circuit. The tip piston actuator resistance shall be 0.59 to 1.03 ohms. Actuator resistance greater than 1.03 ohms shall be classified as "Critical" defect. Actuator resistance less than 0.59 ohms shall be classified as "Major" defect. The base detonator circuit shall exhibit a resistance of 100k ± 10k ohms."

PAGE 5

3.19, Third line, delete specification "MIL-P-70661" and substitute "paragraph 4.5.16".

3.23, Workmanship: Delete paragraph in its entirety and substitute with the following:

3.23 Workmanship.

3.23.1 Metal parts. The TOW/E6 Probe and S&A Assembly and its components shall be fabricated and finished in a thorough, workmanlike manner and shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. Wires shall not be damaged during or after assembly. Solder connections shall be in accordance with paragraph 3.20. The cleaning methods used shall not be injurious to any part nor shall parts be Contaminated by the cleaning agents. All the required marking shall be clearly legible.

* 3.23.2 Plastic parts. Parts shall be free from blisters, cracks, foreign inclusions, scratches, voids or any other surface or internal defect which may detrimentally affect serviceability or appearance, and shall be uniform in color and appearance. Parts may be inspected in the restrained condition. Small amounts of flash, knockout pin marks, gates, sinks and other slight imperfections which do not detrimentally affect serviceability of plastic molded parts need not be selected as inspection points."

3.24, Add new paragraph as follows:

"3.24 Process documentation. Concurrent with submission of the First Article Sample (see paragraph 4.3), the contractor shall submit a written description of the manufacturing process used for the Krytox lubrication of the wedge. The manufacturing process shall be reviewed by and must be acceptable to the Government. The contractor's written plan of process and process controls shall be prepared in accordance with Data Item Description DI-P-1604 (Tailored) (see 6.9).

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Proposed changes to the process, which produced the accepted First Article sample shall be submitted to the Government prior to actual implementation of the change. A new First Article sample may be required for each proposed process change as determined by the Government."

3.25, Add the following new paragraph:

"3.25 Leaf spring contact. The leaf spring contact tip in the safe position shall be inspected in accordance with 4.5.17."

3.26, Add the following new paragraph:

"3.26 Mechanism assembly. The mechanism assembly shall have the shoulder washers and piston actuator spacer present. Inspection shall be performed in accordance with paragraph 4.4.2.61."

PAGE 6

4.3.1, Add at end of paragraph as follows:

"The First Article Sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packing shall be the same as used in regular production and shall be obtained from the same source of supply. A First Article Sample or portion thereof, as Directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as the significantly affect product uniformity or quality as determined by the Contracting Officer."

4.3.2, Line 4: Delete "byu" and substitute "by".

PAGE 41

4.3.2.31, Major 114: Delete paragraph in its entirety.

Major 115: Change paragraph to the following, "Height of tab straight shear, (.015 min.), (see Note 1)".

PAGE 42

4.4.2.31, Minor 208: Delete "3.2" and substitute "3.23".

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PAGE 46

4.4.2.33, Major 124: Delete paragraph in its entirety.
Major 125: Change paragraph to the following: "Height of tab straight shear, (.015 min), (see Note 1)."

PAGE 50

4.4.2.35, Minor 211: Delete "3.2" and substitute "3.23".

PAGE 53

4.4.2.37, Minor 206: Delete "3.2" and substitute "3.23".

PAGE 60

4.4.2.44, Delete Major 101 in its entirety.

Add Major 106 as follows:

"Major	Examination or Test	AQL	Requirement Paragraph	Paragraph Reference
106	Presence of adhesive (see Note 4 of drawing)	100%	3.2	Visual/ gage"

PAGE 76 & 77

4.4.2.60: Next higher assembly: Delete "9398046" and substitute "9388046".

Delete Major 114 in its entirety.

Delete Critical Defect 1 and substitute with the revised critical defect inspections:

"CLASS	EXAM/TEST	CONFORMANCE CRITERIA	REQ'D PARA.	INSP METHOD
Critical	(See note 1)			
1	Modified M114 S&A in safe position	100%	3.2	4.5.1.2a/ 4.5.1.2b/ 4.5.1.2c
2	Tip detonator switch resistance	100%	3.17	4.5.14/STME

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3	Tip actuator circuit resistance	100%	3.18	4.5.15/STME”

Add Major 121 as follows:

“CLASS	EXAM/TEST	CONFORMANCE CRITERIA	REQ’D PARA.	INSP METHOD
Major 121	Presence of shorting screw	100%	3.2	4.5.1.2d”

* Add Major 122 as follows:

“CLASS	EXAM/TEST	CONFORMANCE CRITERIA	REQ’D PARA.	INSP METHOD
Major 122	presence of Silicone tubing	100%	3.2	Visual”

Add the following special inspections:

“ CATEGORY	EXAM TEST	AQL	REQ’D PARA_	PARA REF/ INSP METHOD
Special 10	Absence of adhesive sealant (See Note 18 on Dwg.)	100%	3.2	Visual
20	Torque on screw – 2 places (See Note 3 on Dwg.)	100%	3.2	Gage
30	Depth of housing support screw (See Note 19 on Dwg.)	100%	3.2	Gage”

Delete Note 1 and substitute with the following:

“1. In accordance with MIL-A-48078.”

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PAGE 77a

4.4.2.61, Add new paragraph as included.

Major 101, delete "3.24" under requirement paragraph and substitute with "3.26".

PAGE 77b

4.4.2.62, Add new paragraph as included.

PAGE 79

Add the following note to the Lot Quality Confirmation Test:

"If only one defect occurs during the Breakaway Strength Test, the same defective unit shall be retested by the contractor for the Breakaway Strength Requirement. Lot acceptance can be ranted only when the retest unit meets in full compliance with the test requirements. When retest unit fails to meet the test requirement:

a. Physical Configuration Audit (PCA) on the defective unit shall be performed by the contractor to ensure configuration.

b. Failure analysis report on the defective unit shall be submitted to the Government for review and approval prior to the disposition of the lot."

PAGE 82

4.5.1.2, Delete paragraph in its entirety and substitute with the following paragraph:

"4.5.1.2 Quality conformance.

a. Prior to the assembly of the modified M114 S&A into the probe, visually inspect the safety of the detent spring position as shown in Figure 10.

b. The S&A shall be visually inspected immediately prior to packaging to insure that the rotor is in the safe position (detonator not pointing up).

c. For lot quality conformation testing, the inert S&A shall be visually inspected prior to environmental testing and upon completion of the continuity circuit test to ensure that it conforms to 3.4.

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PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.61	Mechanism Assembly	SHEET 1 OF 1	9388005
			NEXT HIGHER ASSEMBLY 9387141

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Leaf spring in proper position	100%	3.24	Visual
102	Presence of shoulder washers	100%	3.2	Visual
103	Presence of piston actuator spacer	100%	3.2	Visual
Minor				
201	Evidence of poor workmanship	100%	3.23	Visual

NOTES:

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PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.62	Safety and Arming Device Assembly	SHEET 1 OF 1	9231500
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	(See Note 1)			
1	Proper position of rotor shaft	100%	3.4	Visual/(See Fig. 9)
2	Proper gap between spring detent And "G" sensing leaf	100%	3.4	Visual/(See Fig. 9) Visual
Major				
101	Presence of piston actuator	100%	3.4	Visual
102	Absence of safety screw	100%	3.4	Visual
103	Improper gap between spring detent And " G " sensing leaf (one or two Wire diameters)	100%	3.4	Visual/(See Fig. 9)

NOTES:

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d. The S&A shall be visually inspected prior to packaging to insure that the shorting screw is removed (see Figure 2)."

PAGE 84

4.5.7, Add to end of paragraph the following sentence:

"A record of actual deployment time for each probe shall be maintained for the lot quality confirmation testing (LQCT)."

PAGE 88

4.5.10.2, Add the following new paragraph:

"c. Tip assembly Upon insertion of inert charge assembly (Dwg. 9399673), inner tip assembly (Dwg. 9399684), and crush ring (Dwg. 9399727), assemble the tip assembly (Dwg. 9399730) to the probe with 55 + 5 in-lbs torque and cure at ambient for 72 hours minimum before commencing first article or lot quality confirmation testing."

Add Major 104 as follows:

" Major 104 …… Rotor torsion …… 100% …… 3.4 …… 5.18/Visual assembled."

PAGE 90

4.5.14.2.a, After the first sentence add the following:

"This test shall be conducted prior to continuity circuit tests described in 4.5.15 (Table V). The test circuit shall not exceed 40 microamperes."

Second line, replace "2 milliampers" with "40 microampers for live S&As, or 2 milliampers for inert S&As."

Table IV, Delete in its entirety and substitute new Table IV as follows:

"TABLE IV - Open circuit test

	TEST	FROM (+)	TO (-)	RESISTANCE (ohm)
1.	Tip detonator switch	Red cable	Inner tip	≥ 1.0 meg
2.	Tip switch	Inner tip	Black cable	≥ 1.0 meg

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3.	Ogive switch	Base contact	ogive (or rivet)	≥ 1.0 meg

The polarity for the tip detonator switch is reversible."

4.5.14, delete paragraph in its entirety and substitute with:

"4.5.14 Open circuit tests. Test in accordance with Table IV. Due to the capacitive nature of the circuit, steady state shall be achieved before performing actual measurements. (ref. 3.17)"

PAGE 91

Table V, Delete in its entirety and substitute the new Table V as follows:

"TABLE V - Continuity Tests"

TEST	FROM (+)	TO (-)	RESISTANCE (ohm)
Tip ground circuit	Outer tip	Black cable	≤ 1.0
Ogive switch ground circuit	Ogive (or rivet)	Black cable	≤ 1.0
Base S&A actuator circuit	White/yellow cable	Yellow cable	≤ 1.0
Ogive switch live circuit	Base contact	E10	≤ 1.0
Inner tip circuit	Inner tip	T-lug (S&A)	≤ 1.0
Tip detonator circuit	Pin S&A	Red cable	≤ 1.0
Tip actuator firing circuit	White wire (S&A)	White/black cable	≤ 1.0
Tip piston actuator resistance	White/black cable	Black cable	.59 to 1.03
Base S&A detonator	White/red	White/brown	100k +/- 10k

T-lug - terminal lug (dwg. 9388028)
pin (S&A) - electrical contact stud (dwg. 9231433)

4.5.16, sixth line down after the firing energy, add sentence "in accordance with MIL-P-70661".

Add the following new paragraphs:

"4.5.17, Visual examination for proper leaf spring contact position (See Figure 7 and 8). This examination shall be performed 100 percent at the receiving inspection of the M114 S&A devices."

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"4.5.18, Visual inspection for proper installation of rotor torsion spring (See Figure 11). This inspection shall be performed 100 percent following removal of the S&A cover before assembling device into probe. A unit found to be improperly assembled shall be set aside. Authorized Government personnel shall perform an analysis as to cause."

5.2 Packing. Packing of the probe and its associated components shall be in accordance with MIL-STD 1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. The probe and its associated components shall be shipped in the stowed condition in accordance with notes 16 and 17 of Dwg. 9387141."

PAGE 91a

Add Figure 7, included.

PAGE 91b

Add Figure 8, included.

PAGE 91c

Add Figure 9, included.

PAGE 91d

Add Figure 10, included.

PAGE 91e

Add Figure 11, included.

PAGE 92

Add the following:

"3.24 Manufacturing Methods Report DI-P-1604."

6.4, fourth line, delete the following in its entirety:

"3.21 Quality Inspection Test, Demonstration and Evaluation Report, DI-R-1724."

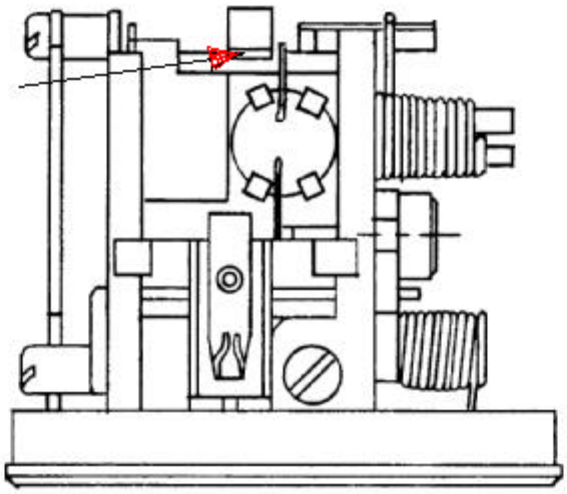
6.9, Add new paragraph as follows:

"6.9 Process documentation (see 3.24

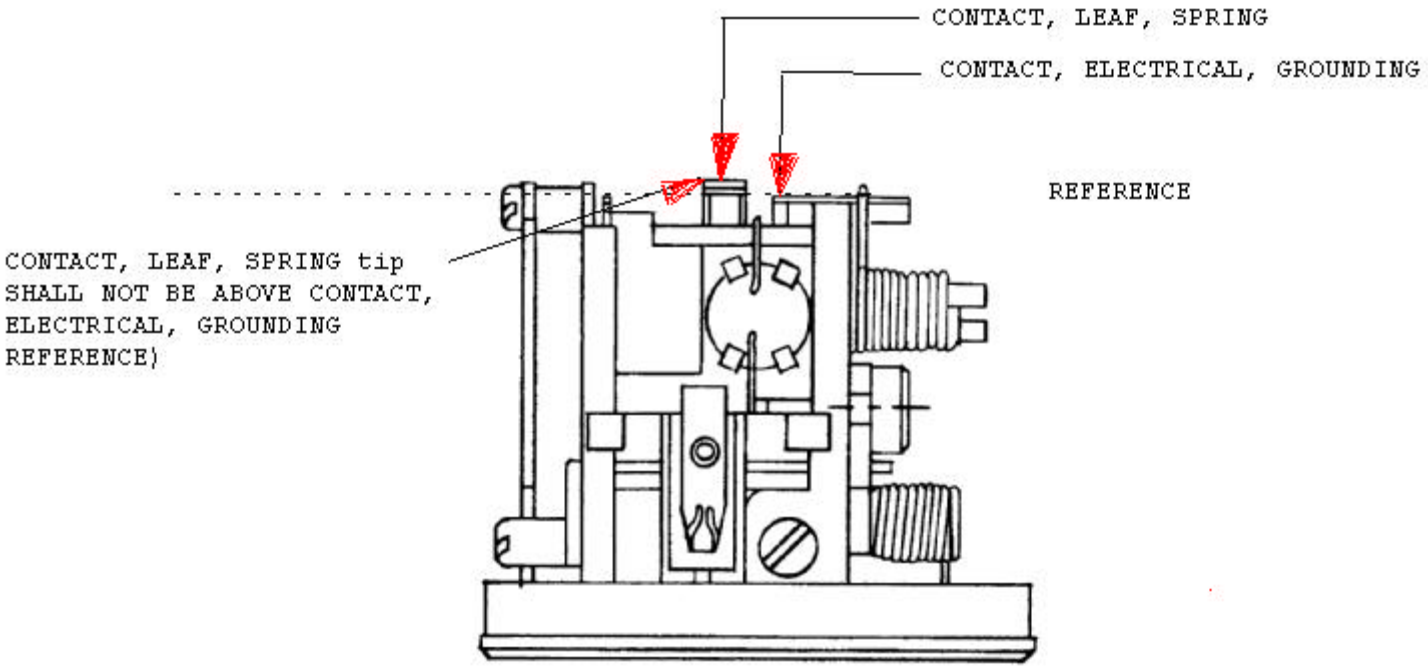
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M114 S&A MECHANISM ASSEMBLY

CONTACT, LEAF, SPRING tip SHALL BE BELOW OR FLUSH WITH CONTACT, ELECTRICAL, GROUNDING (REFERENCE)



ACCEPTABLE CONDITION

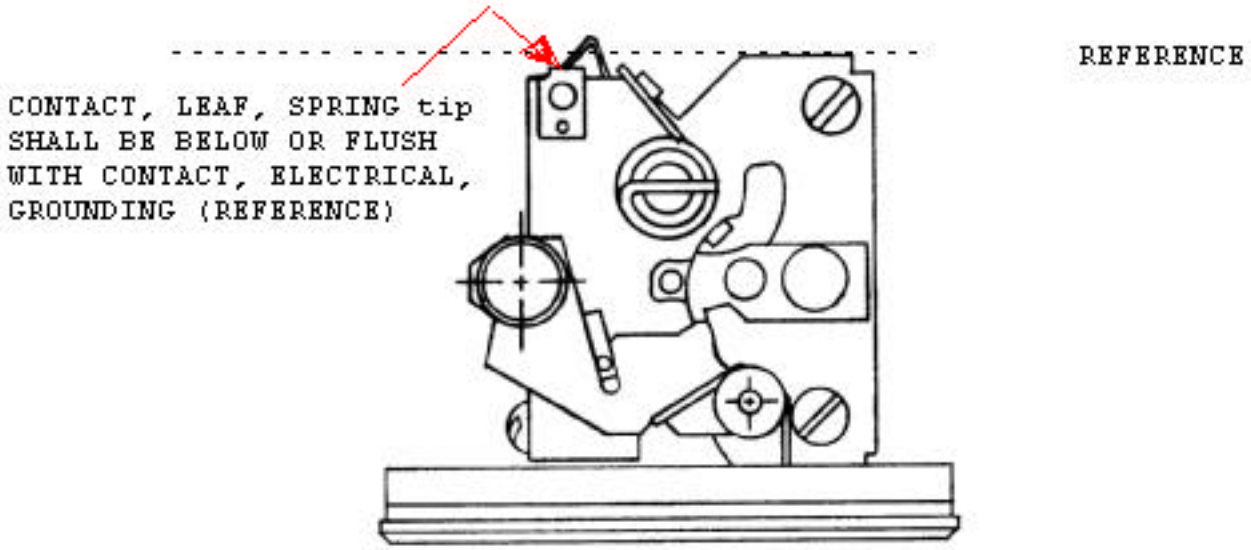


REJECT CONDITION

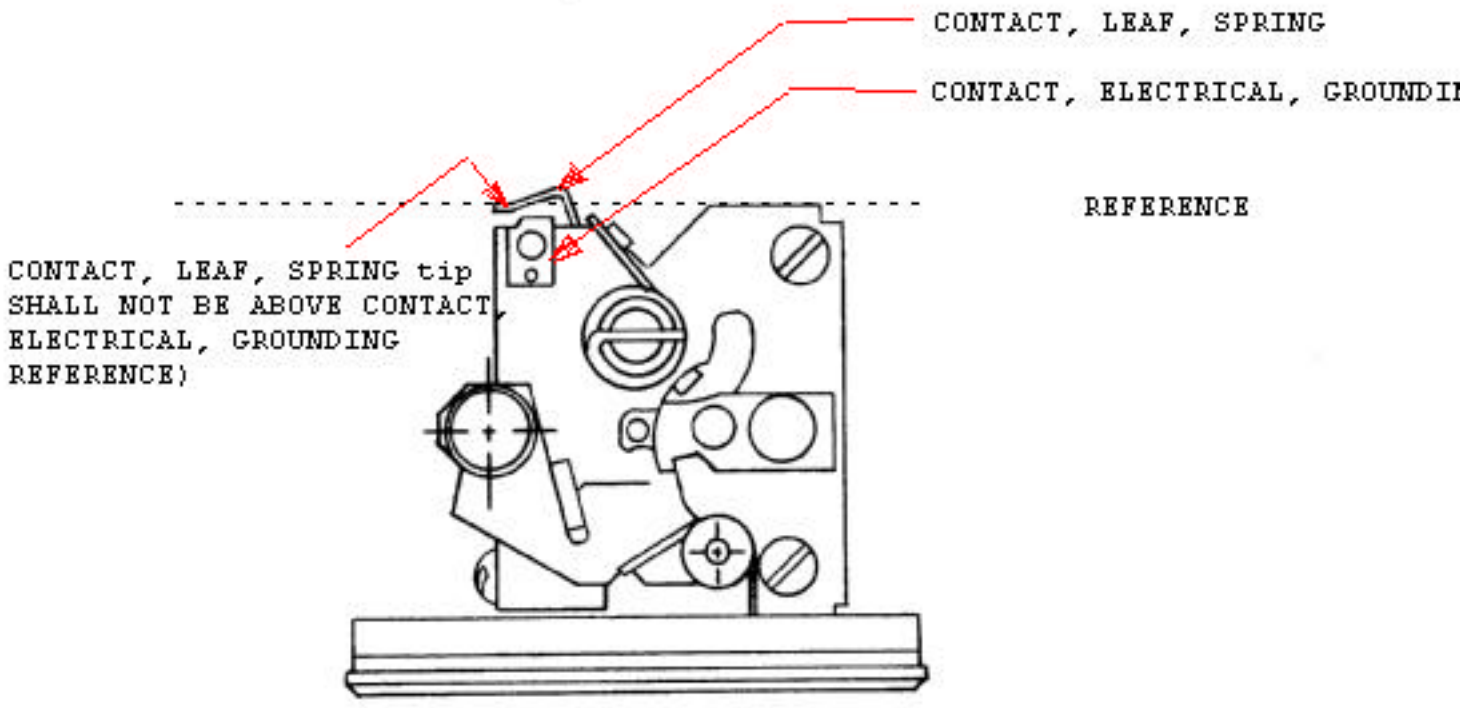
Figure 7. Acceptance criteria for leaf spring contact (front views)

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M114 S&A MECHANISM ASSEMBLY



ACCEPTABLE CONDITION

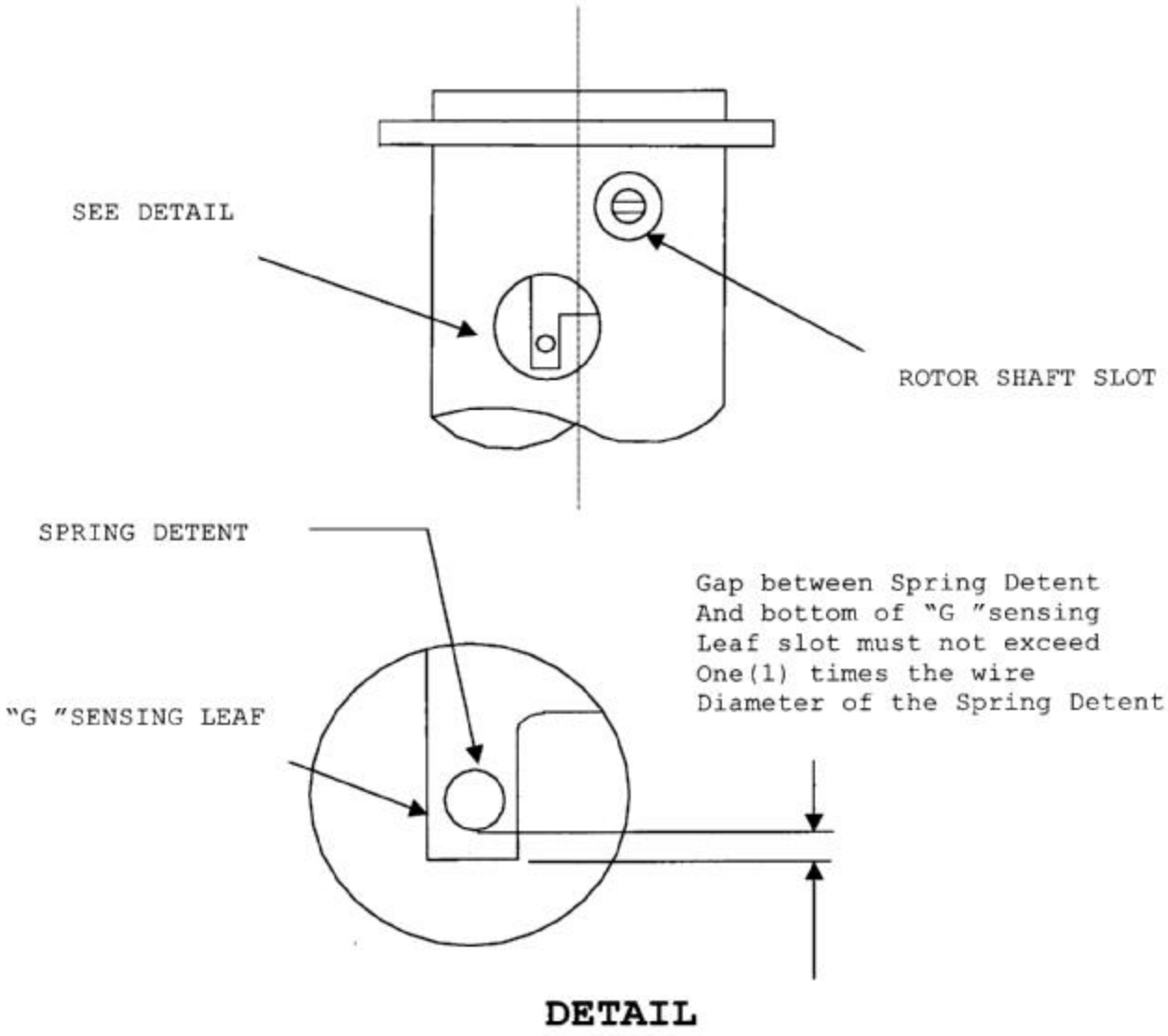


REJECT CONDITION

Figure 8. Acceptance criteria for leaf spring contact (side view)

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SAFETY AND ARMING DEVICE, M114

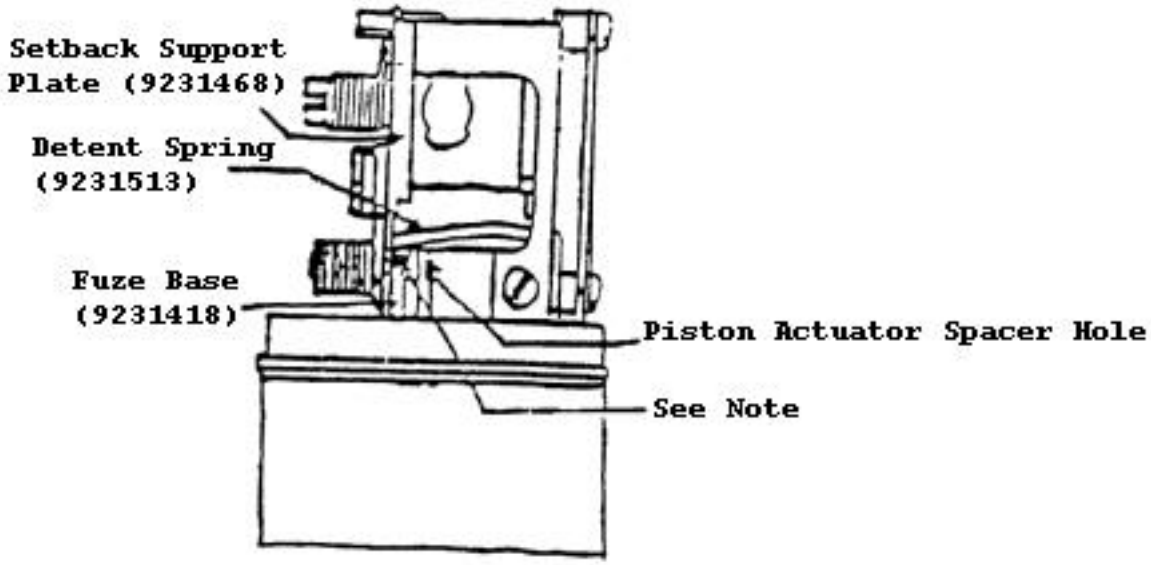


DETAIL

WARNING
Should the S&A fail to be in the safe condition, an extreme safety hazard shall be considered to exist, and the warhead section disposed of in accordance with acceptable procedures

Figure 9. The safe condition of the M114 Safety and Arming Device

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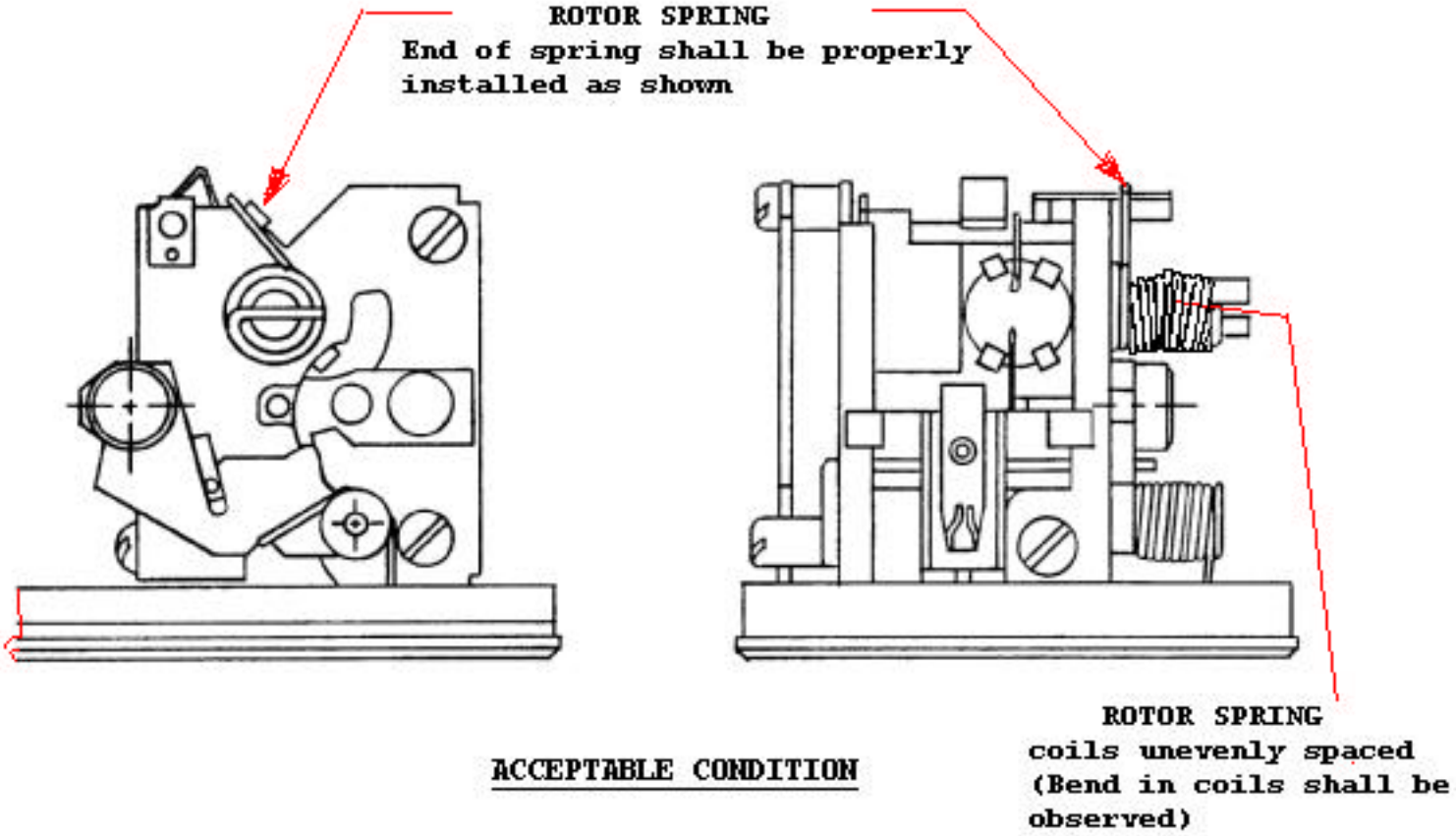


View B - B
Rotated 180°

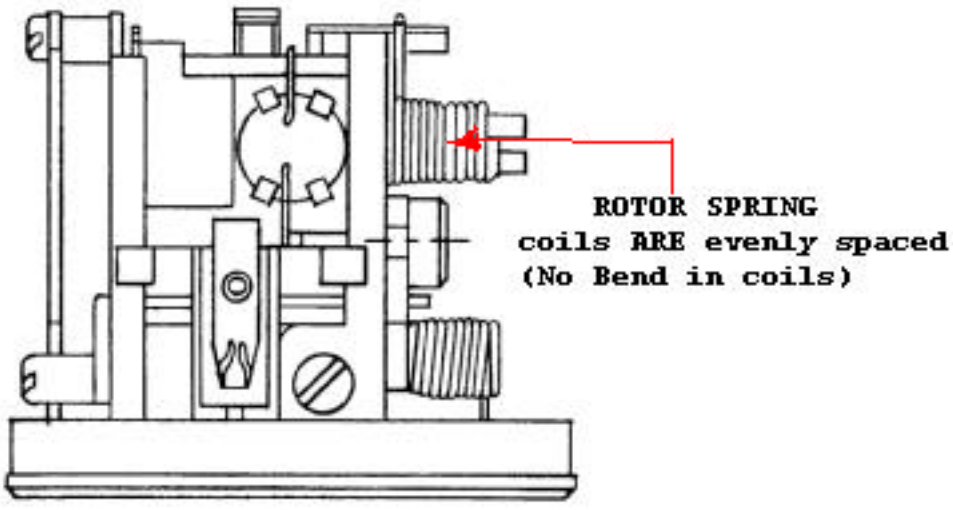
"Note Detent spring (9231513) shall contact the fuze base (9231418) between the piston actuator spacer hole and setback support plate (9231468), as shown.

Figure 10. Safety of the detent spring position

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ACCEPTABLE CONDITION



REJECT CONDITION

Figure 11. Acceptance criteria for Rotor Spring Assembly

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